2025-06-09v0.0.2by Max Karrebæk for use in notes and written material as part of studies associated with BSc in engineering physics at DTU.

The DTUFysIng package*

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June 16, 2025

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1 Usage and examples

1.1 Easing adherence to ISO standards

ISO requires that dimensionless constants (numbers) such as $\sqrt{-1}$ are typeset in roman font whereas physical dimensions are set in an *italiziced* font. Not surprisingly, being strict turns out to be useful eg. as part of an analysis of a mechanical system, you might have introduced the distance d. Later on, it is necessary to multiply the time derivate of some dimension x by this distance. This can be written either incorrectly or the correct way:

$$d\frac{dx}{dt} \quad \text{incorrect and confusing} \tag{1}$$

$$d\frac{\mathrm{d}x}{\mathrm{d}t} \quad \text{correct and understandable} \tag{2}$$

We therefore define the following macros.

`\im` `\e` `\dd` These macros should only be used in math mode. Their use can be seen in table 1. Electrical engineers should redefine the macro `\im` by adding

`a + b\im`	$a + b\mathrm{i}$
`\e^{-x^2}=\exp{(-x^2)}`	$\mathrm{e}^{-x^2} = \exp\left(-x^2\right)$
`\dd x`	$\mathrm{d}x$

Table 1: Roman numbers.

*This document corresponds to DTUFysIng v0.0.2, dated 2025-06-09.

\def\im{\mathrm{j}} to their preamble—maybe this will be a package option in the future.

1.2 Calculus

\diff \pdiff

Lists of differentials can be written as $\diff{x,y,z}$ yielding $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$. This is very useful when doing vector calculus! Consider the following integral.

$$\iiint x+y+z\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \tag{3}$$

This can be written in the compact and readable expression `\iiint x+y+z\diff{x,y,z}` as opposed to writing the integrands explicitly `\,\dd x\,\dd y\,\dd z` or worse, by using `\mathrm{d}`. A corresponding command `\pdiff{x,y,z}` is also supplied: $\partial x\,\partial y\,\partial z$. The interproduct spacing can be modified using an optional argument `\pdiff[]{\alpha,\beta}`: $\partial\alpha\partial\beta$

\dif \pdif

DTUFysIng makes it very easy to write the differential quotient. This can be achieved using either `\dif` og `\pdif`:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2}+\omega^2\frac{\mathrm{d}x}{\mathrm{d}t}=0 \tag{4}$$

Simply, by writing `\dif[x]{2}{t} + \omega^2\dif[x]{}{t}=0`. More complicated expressions are also allowed

$$\frac{\partial^3}{\partial x^2\partial y}f(x,z)=0 \tag{5}$$

`\pdif{3}{x^2,y}f(x,z)=0`

1.2.1 Vectors

\vect

Vectors are defined using `\vect`. Maybe the macro should just redefine the plain command `\vec`. Vectors are set as boldsymbol: `\vect{f}(\vect{x})` yields $\boldsymbol{f}(\boldsymbol{x})$.

\dive \rot \grad

The divergence, rotation and gradient of a vector field can be written as

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{c})=\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{c})-\nabla^2\boldsymbol{c} \tag{6}$$

`\rot{(\rot{\vect{c}})} = \grad{(\dive{\vect{c}})} - \nabla^2\vect{c}`. Divergence is `\dive` and not `\div` as this macro is already defined to be the (useless) symbol $\div$.

2 Implementation

\im \e \dd

These macros should probably be renamed to `\imaginary`, `\euler` and `\differentiald`, for compatibility reasons. The user would then be advised to define shorthand macros: `\newcommand*\im\imaginary` etc. Maybe for v0.0.3.

```
\def\im{\mathrm{i}}%
\def\e{\mathrm{e}}%
\def\dd{\mathrm{d}}%
```

\ddspace `\ddspace` is an internal macro used in differentials. Maybe it should be defined as a length and not just a macro… Another project for v0.0.3!

```
\let\ddspace=\thinspace%
```

\diff \pdiff They are both wrappers for the internal macro `\@iff`. They share the same syntax, `\diff[`⟨*spacing*⟩`]{`⟨*comma-separated list*⟩`}`.

```
\def\@diff#1#2#3{\@for\@iff:={#1}\do{#2#3\@iff}}%
\newcommand*{\diff}[2][\ddspace]{\@diff{#2}{#1}{\dd}}%
\newcommand*{\pdiff}[2][\ddspace]{\@diff{#2}{#1}{\partial}}%
```

This counter will be useful. Alongside some internal macros. It is somewhat arbitrarily set to zero.

```
\newcount\len@diff%
\len@diff=\z@%
```

\@lenlist Internal macro that measures whether a cs-list has one or more entries. If there is one entry the counter `@lenlist` is set to `1` and `2` otherwise. Maybe it should check whether there are entries.

```
\def\@lenlist#1,#2\@nil{%
  \def\@tempa{#2}
  \ifx\@tempa\@empty%
    \len@diff=\@ne
  \else
    \len@diff=\tw@
  \fi%
}%
```

\dif \pdif First, the internal command for differential quotients.

```
\def\@dif#1#2#3#4{%
  \@lenlist#3,\@nil
  \ifnum\len@diff=1
    \def\@tempa{#2}
    \ifx\@tempa\@empty
      \frac{#4#1}{#4#3}
    \else
      \frac{#4^{#2}#1}{#4#3^{#2}}
    \fi
  \else
    \frac{#4^{#2}#1}{\@diff{#3}{}{#4}}
  \fi%
}%
```

`\dif` and `\pdif` command should be used. They feature the same syntax, `\dif[`⟨*function*⟩`]{`⟨*degree*⟩`}{`⟨*denominator, list*⟩`}`. The argument `{`⟨*degree*⟩`}` can be left empty which is normal for first degree differentials: $\frac{\partial f}{\partial x}$ by `\pdif[f]{}{x}`. Of course both commands are purely for use in math mode. Advanced differentials: $\frac{\mathrm{d}^n}{\mathrm{d}x^{n-2}\mathrm{d}y^2}$ by `\dif{n}{x^{n-2},y^{2}}`.

```
\newcommand*\dif[3][]{\@dif{#1}{#2}{#3}{\dd}}%
\newcommand*\pdif[3][]{\@dif{#1}{#2}{#3}{\partial}}%
```

Other differential quotiens could easily be defined.

\vect \rot \dive \grad \del These are all trivial, `f(\vect{r})`: $f(\boldsymbol{r})$.

```
\def\vect#1{\boldsymbol{#1}}%
```

operators

```
34 \newcommand*\rot[1]{\vect{\nabla}\times #1}%
35 \newcommand*\dive[1]{\vect{\nabla}\cdot #1}%
36 \newcommand*\grad[1]{\vect{\nabla} #1}%
37 \newcommand*\del{\mathrm{\Delta}}%
```

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